

SOCIAL NETWORKS

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Week 5 — Question Paper (Comprehension-Based MCQ / MSQ)

Comprehension 1 — How an Open-Plan Office Sorts Itself: The Schelling Model

Case Study

A workplace analyst, Ravi, wants to understand why an open-plan office drifts into team enclaves even though no one is assigned a fixed seat. He models the office as a square grid of hot-desks. Every desk is either free or taken by an employee from one of two teams call them Design and Engineering. A desk's neighbours are the desks touching it, diagonals included, so a desk in the middle of the floor has eight neighbours, a desk in a corner has only three, and a desk along an edge but not at a corner has five.

Each employee has the same modest preference, captured by a threshold t : an employee is content as long as at least t of its neighbouring desks are occupied by teammates. An employee with fewer than t neighbouring teammates is unsatisfied and moves to a randomly chosen free desk. Crucially, when one person moves it changes the neighbourhoods of others, so some previously content employees may become unsatisfied and vice versa; the set of unsatisfied employees therefore has to be recomputed each day. Ravi runs the reseating day after day and watches the overall pattern.

He is careful to separate what the rule forces from what it merely tends to produce. He can hand-arrange mixed seatings in which every employee is content, so clustering is not logically unavoidable yet when employees act on the rule, sharply sorted team blocks appear again and again. He also notices that a higher threshold produces cleaner separation but takes far longer to settle, and that an extreme demand can stop the floor from ever settling at all.

Questions 1–10

Q1. [MCQ] What is the central finding that Ravi's simulation illustrates?

- A. A central authority forces every employee to sit only among its own team.
- B. Team blocks appear only when more than half of the desks are left free.
- C. Each employee follows a mild local preference, yet the floor still sorts itself sharply.
- D. Employees always end up evenly mixed, since random moves cancel any clustering.

Answer: C

Reference: Lecture 56 – Spatial Segregation: Conclusion (≈00:34–01:35)

Solution: The striking point is that no one demands a sorted office; each employee only wants a modest number of nearby teammates. Those mild local choices, repeated by everyone, still drive the whole floor into well-separated team blocks. There is no central planner, no need for free desks to dominate, and the moves do not cancel into an even mix.

Q2. [Match the following] Match each kind of desk in Column A with the correct fact in Column B.

Column A (desk)	Column B (fact)
1. Internal desk (not on the floor's edge)	a. has 3 neighbouring desks
2. Corner desk	b. never makes anyone unsatisfied (no team to be unhappy about)
3. Edge desk (on the edge, not a corner)	c. has 8 neighbouring desks
4. Free (empty) desk	d. has 5 neighbouring desks

- A. 1-a, 2-c, 3-d, 4-b
- B. 1-c, 2-a, 3-d, 4-b
- C. 1-c, 2-d, 3-a, 4-b
- D. 1-c, 2-a, 3-b, 4-d

Answer: B (1-c, 2-a, 3-d, 4-b)

Reference: Lecture 57 – Schelling Model Implementation: Introduction (≈01:24–03:48)

Solution: Counting touching desks including diagonals: an internal desk has 8 neighbours, a corner desk has 3, and a non-corner edge desk has 5. A free desk has no team, so it is never counted as unsatisfied and no one relocates from it.

Q3. [MCQ] An employee is content only if at least half of its eight neighbouring desks are teammates. What threshold t does this correspond to?

- A. $t = 2$
- B. $t = 8$
- C. $t = 4$
- D. $t = 6$

Answer: C

Reference: Lecture 55 – Simulation of the Schelling Model (≈01:30–02:00)

Solution: Half of eight neighbours is four, so demanding 50% teammates sets the threshold at $t = 4$. (For reference, 25% of eight is 2 and 75% of eight is 6.)

Q4. [MCQ] An employee at an internal desk has 8 neighbours, of which 2 are teammates, and the threshold is $t = 3$. Is the employee content, and why?

- A. Content, because 2 neighbouring teammates is more than enough at this threshold.
- B. Content, because only internal desks with 8 neighbours are ever counted at all.
- C. Unsatisfied, because it has 8 neighbours and 8 is well above the threshold of 3.
- D. Unsatisfied, because 2 neighbouring teammates is fewer than the threshold of 3.

Answer: D

Reference: Lecture 57 – Schelling Model Implementation: Introduction (≈03:48–end)

Solution: Contentment depends on the number of neighbouring teammates, not the total number of neighbours. With only 2 teammates against a threshold of 3, the employee falls short and is unsatisfied, so

they move to a free desk. The count of 8 is the neighbourhood size, which is not what the threshold compares against.

Q5. [MCQ] Suppose every employee demands that about 89% of its eight neighbouring desks be teammates. What is the likely outcome?

- A. Almost no seating can satisfy everyone, so people keep reseating and the floor may never settle.
- B. The office sorts faster than at any lower threshold, settling within just a few days of moves.
- C. Everyone is content immediately, since 89% of eight rounds down to only two teammates.
- D. All the free desks disappear at once, because every employee finds a satisfying desk.

Answer: A

Reference: Lecture 55 – Simulation of the Schelling Model (≈05:29–end)

Solution: 89% of eight is about seven, so an employee would need nearly all eight neighbours to be teammates, which almost no seating achieves. People keep moving in search of an impossible desk, and the simulation may never converge. Higher demands sharpen the sorting only up to a point; an extreme demand prevents settling rather than speeding it.

Q6. [MCQ] Which statement about whether the team sorting is inevitable is correct?

- A. Sorting is logically guaranteed; no content seating can ever stay mixed.
- B. Sorting never occurs in the model; employees always stay uniformly mixed.
- C. Sorting is not forced — mixed content seatings exist — yet it usually emerges.
- D. Sorting depends only on the size of the floor, not on employees' thresholds or their choices.

Answer: C

Reference: Lecture 56 – Spatial Segregation: Conclusion (≈01:35–end)

Solution: One can hand-build seatings where everyone meets the threshold without clustering, so sorting is not mathematically unavoidable. Despite that, when employees follow the relocation rule the floor almost always drifts into separated team blocks. It is a strong tendency, not a logical certainty, and it is driven by the employees' choices rather than fixed by floor size.

Q7. [MCQ] Why does the simulation recompute the set of unsatisfied employees each day?

- A. Moving one employee can flip its neighbours' status, so the unsatisfied set changes each day.
- B. Once the unsatisfied list is built it never changes, so it is computed only a single time.
- C. An employee always moves to the desk with the most teammates, never to a random one.
- D. Content employees relocate each day, while unsatisfied employees always stay in place.

Answer: A

Reference: Lecture 60 – Getting a list of unsatisfied nodes; Lecture 61 – Shifting nodes

Solution: Each move rewrites the neighbourhoods around both the vacated desk and the new desk, so employees who were content can become unsatisfied and vice versa. The unsatisfied set is therefore stale after any move and must be rebuilt each day. In the version implemented, the mover goes to a random free desk, and it is the unsatisfied employees who move, not the content ones.

Q8. [MCQ] In this model, what does it mean to say that “local behaviour produces a global pattern”?

- A. A central plan sets the floor-wide pattern that each employee must individually obey.
- B. Each employee can see and control the whole floor's seating before deciding to move.
- C. The global pattern is fixed first, and employees then pick desks to fit that pattern.
- D. Many employees each acting on one simple local rule together yield a floor-wide pattern.

Answer: D

Reference: Lecture 56 – Spatial Segregation: Conclusion (≈00:34–01:35)

Solution: Emergence means the large-scale pattern is not imposed or planned; it arises from many individuals independently following the same simple local rule. No employee sees or controls the whole floor, and the pattern is an outcome of those local moves rather than a blueprint they fit themselves into.

Q9. [MSQ] Which statements about the model's rules are correct? (Select all that apply.)

- I. An employee is content if at least t of its neighbouring desks are teammates.
 - II. A free desk has no team and is never counted as unsatisfied.
 - III. Raising the threshold tends to increase the sorting but slows convergence.
 - IV. A corner desk has 8 neighbours, the same as an internal desk.
- A. I, II, IV
 - B. II, III, IV
 - C. I, II, III
 - D. I, III

Answer: C (statements I, II, III)

Reference: Lectures 55–57 – Schelling model and its implementation

Solution: The threshold rule (I), the inert free desks (II), and the trade-off in which a higher threshold sharpens separation but slows settling (III) are all correct. A corner desk has only 3 neighbours, not 8, so IV is false.

Q10. [MSQ] Which statements about neighbourhoods and emergence are correct? (Select all that apply.)

- I. An internal desk has 8 neighbours, counting the diagonal desks.
 - II. A non-corner edge desk has 5 neighbours.
 - III. The same local reseating rule, applied by everyone, yields floor-wide team sorting.
 - IV. The model proves the sorting is mathematically unavoidable in every case.
- A. I, II, IV
 - B. I, II, III
 - C. II, III, IV
 - D. I, IV

Answer: B (statements I, II, III)

Reference: Lectures 56–57 – Conclusion and implementation

Solution: Internal desks have 8 neighbours and non-corner edge desks have 5 (I, II), and the shared local rule produces floor-wide sorting (III). The model shows the sorting is a strong tendency, not a logical certainty — content mixed seatings exist — so IV is false.

Comprehension 2 — Allies and Rivals: Structural Balance in a Reality-Show Cast

Case Study

A data analyst, Sara, studies the web of relationships among the contestants of a reality competition. Every pair of contestants has a standing relationship that is either an alliance (a positive tie) or a rivalry (a negative tie), and since the cast is small, every pair has some relationship so the contestants form a complete signed graph. Sara is interested in which local arrangements sit comfortably and which generate pressure to change.

She focuses on triangles of three contestants. A triangle of three mutual allies is comfortable, and so, perhaps surprisingly, is a triangle of two allied contestants who share a common rival the shared opponent only reinforces their bond. By contrast, a contestant allied to two others who are themselves rivals is under strain, and a triangle of three mutual rivals is also unstable, since two of them are apt to join forces against the third. A triangle is comfortable exactly when an odd number of its three ties are positive.

When unstable triangles are present, the cast tends to rearrange itself ties flip until every triangle is comfortable. Sara proves a clean structural result about where this lands, and then implements the whole process: she builds a random complete signed graph, lists its triangles, classifies each, repeatedly repairs an unstable triangle, and finally sorts the settled cast into alliances. The key building block is `itertools.combinations`, used to enumerate the triangles and, where needed, the edges.

Questions 1–10

Q1. [MCQ] Sara finds three contestants where X is allied with both Y and Z, but Y and Z are rivals. What does the theory predict?

- A. The configuration is stable and will persist unchanged for the foreseeable future.
- B. It is unstable; the tension tends to resolve — e.g. Y and Z ally, or X drops one tie.
- C. It is unstable, and the only way it can resolve is all three becoming mutual rivals.
- D. It is stable, because every contestant in the triangle has at least one ally present.

Answer: B

Reference: Lecture 63 – Structural Balance (≈01:57–04:38)

Solution: Two allies of X who are rivals of each other put X in a bind — this is the unstable two-positive, one-negative triangle. The strain typically resolves either by the two rivals becoming allies or by X breaking one alliance. Becoming three mutual rivals is not the forced outcome, and the presence of an ally does not make the triangle stable.

Q2. [MCQ] What is the rule that decides whether a triangle of signed ties is stable?

- A. A triangle is stable when it has an odd number of positive ties (one or three).
- B. A triangle is stable when it has an even number of positive ties (zero or two).
- C. A triangle is stable only when all three of its ties are positive (alliances).
- D. A triangle is stable only when all three of its ties are negative (rivalries).

Answer: A

Reference: Lecture 64 – Enemy's Enemy is a Friend (≈06:12–end); Lecture 68

Solution: A signed triangle is balanced exactly when an odd number of its ties are positive: three positives (all allies) or one positive (two allies with a shared rival). The even cases — two positives or zero positives — are the unstable ones. Requiring all-positive or all-negative is too narrow.

Q3. [MCQ] Which scenario is stable, and which social rule does it follow?

- A. Three mutual rivals are stable, because a shared grudge keeps all of them apart.
- B. Two allies with a common rival are unstable and must add a third rivalry.
- C. Two rivals with a common ally are stable, since the shared ally binds them together.
- D. Two rivals who share a third common rival tend to ally — enemy's enemy is a friend.

Answer: D

Reference: Lecture 64 – Enemy's Enemy is a Friend (≈01:37–end)

Solution: When two contestants are each rivals of a third, they have a common opponent, and that shared enmity tends to push them into an alliance — the “enemy's enemy is a friend” pattern, which yields a stable one-positive triangle. Three mutual rivals are unstable, and two allies with a common rival are already stable (not in need of repair).

Q4. [MCQ] According to the balance theorem, what does a fully stable complete signed network look like?

- A. It always splits into exactly three alliances that are mutually hostile to one another.
- B. It is either one all-allied cast, or two alliances: allied within each, hostile across them.
- C. It must have every contestant allied with every other, with no rivalries present anywhere.
- D. It can split into any number of alliances, with the count depending on the cast's size.

Answer: B

Reference: Lecture 66 – Balance Theorem; Lecture 67 – Proof of Balance Theorem

Solution: The balance theorem says a balanced complete signed network has just two possible shapes: a single bloc of mutual allies, or two alliances with friendships inside each and rivalries between them. It is never three or more alliances, regardless of size, and it need not be rivalry-free.

Q5. [MCQ] Sara models 6 contestants as a complete signed graph. How many triangles must she check for stability?

- A. 6, one triangle for each contestant in the cast.
- B. 15, one triangle for each pair of contestants.
- C. 30, counting the ordered triples of contestants.
- D. 20, one for each unordered triple of the contestants.

Answer: D

Reference: Lecture 69 – Outline of implementation (≈02:25)

Solution: A triangle is an unordered choice of 3 contestants, so the count is 6 choose 3 = 20. There are 15 pairs (which give edges, not triangles), and ordering the triples would overcount each triangle six times, so neither 6, 15, nor 30 is right.

Q6. [MCQ] In the implementation, how are the triangles obtained from the list of contestants?

- A. combinations(nodes, 2) lists every pair, and each pair is treated as one triangle.
- B. combinations(nodes, 3) lists every triple, and each triple is one triangle to classify.
- C. permutations(nodes, 3) lists the triangles, so each triangle shows up six separate times.
- D. combinations(nodes, 3) returns the edges, whose signs are then added together.

Answer: B

Reference: Lecture 70 – Creating graph and counting unstable triangles (≈10:15–18:00)

Solution: Each triangle is an unordered triple of nodes, so combinations(nodes, 3) enumerates them with no repeats. permutations would list each triangle six times, and combinations returns triples of nodes (the triangles), not the edges.

Q7. [Arrange in order] Arrange the steps of the structural-balance implementation in the correct order.

Steps (in scrambled order):

- (i) Repeatedly pick an unstable triangle and flip a tie to make it stable, until none remain.
 - (ii) Build a complete signed graph, assigning each tie + or – at random.
 - (iii) Split the now-stable cast into two alliances (allies together, rivals apart).
 - (iv) Enumerate every triangle with combinations(nodes, 3) and record its three signs.
 - (v) Count how many of the triangles are unstable.
- A. ii → v → iv → i → iii
 - B. iv → ii → v → i → iii
 - C. ii → iv → v → i → iii
 - D. ii → iv → i → v → iii

Answer: C (ii → iv → v → i → iii)

Reference: Lectures 69–71 – Outline, counting unstable triangles, moving to stable

Solution: First build the random signed graph (ii), then list every triangle and its signs (iv), then count the unstable ones (v). Next repeatedly repair an unstable triangle until none are left (i), and finally divide the settled cast into the two alliances (iii).

Q8. [MCQ] In the code, a triangle's three signs have been collected into a list. How is it judged unstable?

- A. If the list of signs contains zero or two + marks, the triangle is counted as unstable.
- B. If the list of signs contains one or three + marks, the triangle is counted as unstable.
- C. The triangle is unstable whenever its list of signs contains at least one – mark anywhere.
- D. The triangle is unstable only when all three of the signs in the list are + marks.

Answer: A

Reference: Lecture 70 – Creating graph and counting unstable triangles (≈20:00–23:27)

Solution: The code counts the + signs: zero or two positives marks the triangle unstable, while one or three positives marks it stable. A single negative does not by itself make a triangle unstable (one-positive triangles are stable), and an all-positive triangle is the most stable case of all.

Q9. [MCQ] How does the implementation split a stabilised cast into the two alliances?

- A. Pick a node; put its rivals in its own group and its allies into the other group.
- B. Pick a node; put its allies in its own group and its rivals in the other, then repeat outward.
- C. Pick a node; put every other node into the same single group regardless of the signs.
- D. Pick a node; sort the others into three groups according to their distance from it.

Answer: B

Reference: Lecture 69 – Outline of implementation (≈05:24–end); Lectures 72–73

Solution: Starting from any contestant, its allies join its own alliance and its rivals go to the other; then each newly placed contestant is processed the same way, spreading outward until all are assigned. Putting allies and rivals the wrong way round, lumping everyone together, or using three distance-based groups all contradict the two-alliance structure.

Q10. [MSQ] Which statements about structural balance are correct? (Select all that apply.)

- I. A complete graph on n contestants has about $n^2/2$ signed ties.
 - II. The proof picks one contestant and sorts the rest into its allies and its rivals.
 - III. However many contestants there are, a balanced cast forms at most two alliances.
 - IV. A triangle with exactly two positive ties is stable.
- A. I, II, IV
 - B. II, III, IV
 - C. I, III, IV
 - D. I, II, III

Answer: D (statements I, II, III)

Reference: Lectures 65–67 – Characterising balanced networks and the proof

Solution: A complete graph on n contestants has $n(n-1)/2 \approx n^2/2$ ties (I); the proof fixes one contestant and partitions the rest into its allies and rivals (II); and a balanced cast has at most two alliances however large it is (III). A two-positive triangle is one of the unstable cases, so IV is false.